

CS 131 — Concept Cheat Sheet + Grep Lab

Slides #01–#05 and Git, in the clean app layout, plus the grep & regex lab.

Slide 01 Intro & Remote Tools

BIG PICTURE

goal	Command-line tools to process big data on remote Linux servers
server	A shared multi-user Linux box (the DGX) — many users at once
tools	Git (version control) and SSH (Secure Shell) to reach it
SSH	
SSH	Secure Shell — encrypted login to run commands on a remote machine
login	ssh user@host opens a remote shell session

Slide 02 Linux Command Basics

FILESYSTEM & PATHS

structure	Files form a hierarchical (tree) structure; the top is root /
absolute	Starts from root / — e.g. /home/me/cs131
relative	Starts from the current working directory
.	The current working directory
..	The parent directory
~	Your home directory

NAVIGATE & EXPLORE

pwd	Print the current working directory
cd	Change directory (cd .. up, cd ~ home, cd - previous)
ls	List directory contents
file	Report a file's type (not its contents)
cat / less	View file contents (less pages through)
man cmd	Open the manual for a command's options

LS -L & OPTIONS

ls -l	Long format: type+perms, links, owner, group, size, date, name
-a	Show all, including hidden (dot) files
-h	Human-readable sizes (K, M, G)
-r -t -S	Reverse order / sort by time / sort by size
combine	ls -alhr = all + long + human + reversed

FILE MANIPULATION

cp -r	Copy a directory and its contents
mv old new	Move OR rename
mkdir -p	Make directories (and any missing parents)
rm -ri	Remove recursively, prompting first (no undo!)
ln -s	Create a symbolic link

WILDCARDS (GLOBBING)

* ?	Any number of chars / exactly one char
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[abc]	Any one listed; [^abc] any one NOT listed
[0-9]	A range of characters
examples	Data?.csv (not Data10) · a[0-9][0-9].txt

Slide 03 Text, Redirection & Regex

VIM EDITOR

modes	i insert · Esc normal · :w save · :q quit · :wq :q!
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TEXT TOOLS

cat sort	Concatenate / sort lines (-n numeric, -r reverse)
uniq -c	Count adjacent duplicates (needs sort first)
grep wc	Match a pattern / wc -l count lines
head tail	First / last lines (-n N) · echo print

I/O REDIRECTION

> >>	stdout to a file: overwrite / append
2> &>	stderr to a file / BOTH streams to one file
<	stdin from a file · 2>/dev/null discards errors

FILE DESCRIPTORS

0 1 2	stdin / stdout / stderr
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PIPELINES

 	stdout of one command → stdin of the next
> vs 	> connects command→file; connects command→command

EXPANSIONS & QUOTING

\$(())	Arithmetic — \$((2+3)) = 5
{ }	Brace — {1..5} , file{a,b}
\$VAR	Parameter; \${cmd} = command substitution
quotes	".." keep \$ \ ` ' . ' . ' all literal · \ escape one

Slide 04 Permissions & Processes

SECURITY COMMANDS

id	Show your identity — uid, gid, and groups
chmod	Change a file's mode (permissions) — owner or root only
umask	Set the DEFAULT permissions for new files
su -	Login shell as another user (needs THEIR password)
sudo	Run one command as root, using YOUR password (restrictable)
chgrp chown	Change a file's group / owner (chown needs root)

READING & SETTING PERMISSIONS

string	-rwxr-xr-- = type owner group other
r w x	read=4 · write=2 · execute=1 (add per section)
octal	644= rw-r--r-- 755= rwxr-xr-x 640= rw-r-----
symbolic	u g o a + - = — e.g. chmod u+x f
umask 022	→ new files 644, new dirs 755

PROCESSES & JOBS

process	A running program; each gets a PID · init = PID 1
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ps top	List processes / live monitor · jobs = this shell
cmd &	Run in the background · fg %1 foreground
Ctrl-Z	Suspend · bg %1 resume · disown detach
SIGNALS	
kill	Send a signal (default SIGTERM 15) · kill -9 SIGKILL
Ctrl-C	SIGINT · SIGHUP = terminal closed → kills its jobs

Slide 05

Environment & Utilities

THE SHELL ENVIRONMENT

vars	Shell variables + environment variables
cmds	printenv · export V=x · alias ll='ls -l'

STARTUP FILES

login	Prompted user+pw (SSH) → /etc/profile , ~/.bash_profile
non-login	GUI terminal → ~/.bashrc
hidden	.name (shown by ls -a) · reload source ~/.bashrc

CRON / FIND / DIFF / CURL

cron	crontab -e · min hour dom mon dow cmd · * = every
find	find P -name '*.x' -size +1M (units c/k/M/G)
diff curl	Compare two files / curl -o file URL

Git

Git Version Control

EVERYDAY WORKFLOW

basics	clone status add commit -m push pull
log	git log --oneline = compact history
fetch	Download only (pull = fetch + merge)

BRANCHING & MERGING

switch -c	Create AND switch to a branch · merge name
diff	git diff main feature · delete git branch -d

FIXING MISTAKES

revert	git revert HASH — safe undo (keeps history)
amend	git commit --amend · git rm --cached + .gitignore

MERGE CONFLICTS

markers	<<<<<< ===== >>>>>> → edit, remove, add + commit
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Lab

Grep & Regex Lab

HOW GREP READS A LINE

- 1 One line at a time** — grep checks every line on its own; if the pattern is found anywhere, it prints the whole line.
- 2 Substring by default** — a pattern can match anywhere; **cat** also matches "concatenate". Pin it with **^** (start) and **\$** (end).
- 3 Literal vs special** — the metacharacters **. ^ \$ * [] \ + ? { } | ()** have powers; put **** in front to make one literal (**\.** = a real dot).
- 4 BRE vs ERE** — plain **grep** treats **+ ? { } | ()** as text; **grep -E** makes them operators. Rule of thumb: use **-E**.

5 A line is a record — in the Yelp JSON each line is one record, so a line match = a record match. Field order varies, so anchor on **"field":**.

6 The flags you need — **-i** ignore case · **-c** count · **-n** line numbers · **-v** invert · **-o** only the match · **-E** extended.

REGEX TOKEN REFERENCE (GREP -E ASSUMED)

.	Any single character	.at → hat, cat, bat
*	Zero or more of the previous	ab* → a, ab, abb
+ ?	1+ / 0-or-1 of the previous	ab+ colou?r
^ \$	Anchor start / end of line	^Error done\$
[abc]	Any ONE character listed	[hc]at → hat, cat
[^abc]	Any ONE character NOT listed	[^0-9] non-digit
[a-z]	A range of characters	[A-Z] uppercase
[[:digit:]]	POSIX class (in brackets)	a[[:digit:]]b → a0b
{n} {n,m}	Exactly n / between n and m	[0-9]{3} 3 digits
a b	Alternation (OR)	cat dog
()	Group items together	(ab)+ → ab, abab
\	Escape — make next char literal	\. dot \\$ dollar

WORKED EXAMPLES

^[hc]at\$	Exactly hat or cat (anchored — NOT chat)
a[[:digit:]]b	a0b, a1b, ... a9b
[0-9]{3}-[0-9]{3}-[0-9]{4}	A phone number NNN-NNN-NNNN
"stars":(4\.	4.5 or 5.0 (escape the dot!)
5 5\.0)	
^[^,]+,222,	222 in the 2nd CSV column
\w+@\w+\.\w+	An email address